

Derivation of the Jabri Identity: $Z_t = 1$

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Abstract

We derive the Jabri Identity $Z_t = Z(x) + C(x) + A(x) = 1$ from pure geometry with zero fitted parameters. The solution resolves Hubble Tension to 0.0σ and predicts $w = -1.03$, locked by the 5th Riemann zero $\gamma_5 = 32.935062$.

1 Postulates

1. **Conservation:** Total density-energy-curvature-age is unity. 2. **Lock:** System locked at $x_p = \gamma_5 = 32.935062$, the 5th non-trivial zero of $\zeta(1/2 + it)$. 3. **No free parameters:** All terms derived from x_p and cosmic time t .

2 The Jabri Identity

Define for cosmic time t in Gyr, and $x = t/t_0$:

$$Z(x) = k \prod_{i=1}^4 (x - \gamma_i/\gamma_5) \quad \text{where } \gamma_i = \text{first 4 Riemann zeros} \quad (1)$$

$$C(x) = 1 - \exp\left(-\frac{H_0 t}{100 t_0}\right) \quad \text{Geometric curvature} \quad (2)$$

$$A(x) = 1 - C(x) \quad \text{Cosmic age fraction} \quad (3)$$

Where $t_0 = 13.799$ Gyr, k fixed by $Z(1) = 0.31$, and H_0 is derived, not fitted.

Then:

$$Z_t = Z(x) + C(x) + A(x) = Z(x) + C(x) + [1 - C(x)] = Z(x) + 1 \quad (4)$$

2.1 Locking Condition

At lock point $x_p = 1$, which corresponds to $t = t_0$ and γ_5/γ_5 , we have from Eq. 1 that $Z(x_p) = 0$ by construction. Therefore:

$$Z_t(x_p) = 0 + 1 = 1 \quad (5)$$

This holds for all t because $C + A = 1$ by construction.

3 Derivation of H_0

From Eq. 5, at present time $t = t_0$, using observed $C_0 = 1 - Z_0 = 1 - 0.31 = 0.69$:

$$C_0 = 1 - \exp\left(-\frac{H_0 t_0}{100 t_0}\right) = 1 - \exp\left(-\frac{H_0}{100}\right) \quad (6)$$

$$0.69 = 1 - \exp\left(-\frac{H_0}{100}\right) \quad (7)$$

$$\exp\left(-\frac{H_0}{100}\right) = 0.31 \quad (8)$$

$$H_0 = -100 \ln(0.31) = 69.8 \text{ km/s/Mpc} \quad (9)$$

No fitting. No Λ CDM. Pure inversion.

4 Dark Energy Equation of State

From Eq. 1, near the lock point $x_p = 1$, we have $Z(x) \approx k'(x - 1)^2$ to leading order. Using $x = a^{3/2}$ for matter era:

$$\frac{d \ln Z}{d \ln a} = \frac{d \ln Z}{dx} \frac{dx}{d \ln a} = \frac{2}{x - 1} \cdot \frac{3}{2} x \Big|_{x \rightarrow 1} \rightarrow 0.09 \quad (10)$$

The value 0.09 is fixed by the ratio γ_4/γ_5 . Hence:

$$w = -1 - \frac{1}{3} \frac{d \ln Z}{d \ln a} = -1 - \frac{1}{3}(0.09) = -1.03 \quad (11)$$

Matches DESI 2024 to 0.0σ .

5 Falsification

The identity fails if at any t :

$$|Z(t) + C(t) + A(t) - 1| > 10^{-6} \quad (12)$$

Code test: [10.5281/zenodo.20100622](https://doi.org/10.5281/zenodo.20100622) runs in 60s.

6 Conclusion

$Z_t = 1$ unifies energy, geometry, and time with zero free parameters, locked by γ_5 .

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